# SUMMARY TABLE: MATHEMATICS GRADE 10

*Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: MATHEMATICS GRADE 10 | |
|---|---|
| **Sub-Strand** | Sub-Strand 2.1: Similarity and Enlargement |
| **Driving Question** | DRIVING QUESTION: Why does resizing a passport photo change its size but never its shape — and what is the mathematics that makes this possible? |

| INSTRUCTIONS |
|---|
| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title | What did I observe? | What did I learn? | How does this explain the phenomenon? |
|---|---|---|---|
| **Lesson 1**<br>**The Anchoring Phenomenon — What Happens When We Resize a Passport Photo?** | Students examined two passport photos side by side — a standard 35 mm × 45 mm Kenyan passport photo and an enlarged version — and noticed that the face looks identical in shape but clearly larger. Teacher should prompt with: 'What do you notice? What do you wonder?' and record all responses on the Driving Question Board (DQB). Students should observe that the outline, proportions, and facial features appear unchanged even though the overall photo is bigger. Pedagogical note: Do NOT rush to explain at this stage — the goal is to surface prior knowledge and generate genuine curiosity. Expect misconceptions such as 'the photo is stretched' or 'one side grows more than the other'; log these for later correction. | Students learned that resizing a passport photo produces a figure that looks the same but is a different size — and that this is not accidental. They were introduced to the term 'enlargement' and began to ask: what rule governs this process? The class co-constructed an initial driving question and populated the first DQB entries. Teacher note: Ensure every student contributes at least one question or observation to the DQB; this builds ownership of the inquiry. | The teacher introduced the anchoring phenomenon: a Nairobi photographer uses software to print passport photos at different sizes — 35 mm × 45 mm for a standard application and 70 mm × 90 mm for a visa booklet. Both photos show the same face with no distortion. The teacher explained that over the next 12 lessons, the class will uncover the precise mathematical rules that make this possible. Teacher should connect this to real-life relevance: Kenyan students applying for the KCSE certificate, national ID, or passport all encounter standardised photo sizes. |
| **Lesson 2**<br>**The Passport Photo Puzzle — Discovering the Linear Scale Factor** | Students measured corresponding lengths on the small (35 mm × 45 mm) and large (105 mm × 135 mm) passport photos — width, height, ear-to-ear distance, eye-to-chin distance — and recorded each pair in a table. They then calculated the ratio (large ÷ small) for each pair. Teacher should | Every pair of corresponding lengths produced the same ratio: 3. Students learned that this constant is called the Linear Scale Factor (LSF) and is calculated as LSF = image length ÷ object length. An LSF > 1 indicates enlargement; LSF < 1 indicates reduction. Teacher note: | The teacher formalised the observed pattern: the photographer's software multiplied every single length in the original photo by the same number (3), producing the enlarged image. This is the definition of the Linear Scale Factor. Teacher should write on the board: LSF = Image Length / |

| | | | |
|---|---|---|---|
| | circulate and ensure students are measuring at least four corresponding lengths. Cold-call at least three groups to share their ratio values. Allow 2 minutes wait time before cold-calling. Pedagogical note: Expect slight variation due to measurement error; use this as a teachable moment about precision and rounding. | Emphasise that the LSF applies to EVERY corresponding length simultaneously — this is the mathematical reason why shape is preserved. | Object Length, and demonstrate with at least two worked examples using the passport photo measurements collected by students. Connect back to DQB: 'Which of our earlier questions does this answer?' Update the DQB together. |
| **Lesson 3**<br>**Identifying Properties of Similar Figures: Angles, Ratios, and the Rule Behind the Resize** | Students were given two pairs of polygons (one pair of quadrilaterals, one pair of triangles drawn on grid paper with Kenyan-themed labels — e.g., a plot of land in Kisumu and its scaled map version). They measured all interior angles with a protractor and calculated the ratio of each pair of corresponding sides. Teacher should ask: 'What pattern do you see in the angles? What pattern do you see in the ratios?' Allow 3 minutes for group discussion before cold-calling four students. | Students learned the two-part definition of similar figures: (1) all corresponding angles are equal, AND (2) all ratios of corresponding sides are equal (this constant ratio is the LSF). Both conditions must hold simultaneously. Teacher note: A common misconception is that equal angles alone are sufficient — use a rectangle vs. a square counterexample to show this is not the case. | The teacher connected the two properties back to the passport photo: the camera lens preserves all angles (no distortion of facial features) while multiplying all lengths by the LSF. Teacher should state formally: 'Two figures are similar (~) if and only if corresponding angles are equal AND corresponding sides are in the same ratio.' Write the similarity statement using correct notation (e.g., $\triangle ABC \sim \triangle DEF$) and explain what the ordering of vertices communicates about corresponding pairs. |
| **Lesson 4**<br>**Discovering Similar Figures: Equal Angles and Constant Ratios** | Learners worked with two more figure pairs — a pair of pentagons (representing two sizes of a Maasai beadwork pattern) and a pair of right-angled triangles. For each pair they: (1) measured all interior angles, (2) measured all side lengths, and (3) computed each ratio of corresponding sides. Teacher should require students to record findings in a structured two-column table (Object \| Image) before drawing conclusions. Circulate and check at least five tables for accuracy. | Students confirmed and consolidated: in both figure pairs, (1) all corresponding angles were equal, and (2) all ratios of corresponding sides were equal and constant. Students also practised writing correct similarity statements with proper vertex correspondence. Teacher note: Stress that the ORDER of vertices in the similarity statement is not arbitrary — it encodes which angles and sides correspond. | The teacher used findings from both figure pairs to reinforce the formal definition of similarity and to distinguish similarity from congruence (congruence is similarity with LSF = 1). Teacher should pose the diagnostic question: 'Could two figures have equal corresponding angles but NOT be similar?' Guide students to discover that for triangles, equal angles are sufficient (AA similarity), but for quadrilaterals and above, both conditions must be verified. Update DQB accordingly. |
| **Lesson 5**<br>**Linear Scale Factor and Lengths — Explaining the Photographer's Rule** | Students were given a set of cards showing similar figure pairs (including one passport-photo pair and two map-scale problems set in Kenya — e.g., a road between Nairobi and Nakuru shown at two map scales). For each card they identified the LSF, then used it to calculate one missing length. Teacher should use a card-sort structure: students first sort cards into 'enlargement' (LSF > 1) and 'reduction' (LSF < 1) piles before calculating. Allow 4 minutes for sorting; cold-call three students to justify their sorting decisions. | Students formalised the LSF as the single constant ratio of any image length to its corresponding object length. They practised three calculation types: (a) given both lengths, find LSF; (b) given LSF and object length, find image length; (c) given LSF and image length, find object length. Teacher note: Type (c) is the most error-prone — students often multiply instead of dividing. Anticipate this and prepare a targeted mini-example. | The teacher drew explicit connections between the LSF formula and proportional reasoning students have used since primary school. Show that image length = LSF × object length is a direct proportion relationship. Connect to the passport photo: 'The photographer doesn't measure every feature — the software applies one multiplier to everything. That multiplier is the LSF.' Teacher should also introduce the notation k for LSF and confirm that k = 3 for the driving question scenario. |

| | | | |
|---|---|---|---|
| **Lesson 6**<br>**Calculating and Applying the Linear Scale Factor (LSF) to Find Unknown Lengths** | Students completed a structured problem set of six problems of increasing difficulty, all set in Kenyan contexts: measuring a shamba (farm plot) near Kericho, scaling a diagram of the KICC building in Nairobi, and finding missing side lengths in similar triangles on a map of Lake Victoria's shoreline. Teacher should use a 'show-boards' protocol: each student writes their answer on a mini-whiteboard and holds it up simultaneously so the teacher can assess the whole class at a glance before revealing the correct answer. | Students learned to fluently apply LSF = image length ÷ object length in all three calculation directions. They also learned to identify corresponding sides correctly before applying the formula — a prerequisite step that, if skipped, leads to errors. Teacher note: Require students to always write a correspondence statement (e.g., AB corresponds to PQ) before setting up any ratio. | The teacher modelled a full worked solution with explicit annotation of each step: (1) identify similar figures and write similarity statement; (2) identify the known corresponding pair; (3) calculate LSF; (4) apply LSF to find unknowns; (5) state answer with units. Emphasise that the LSF is dimensionless (it is a ratio of lengths, so units cancel). Connect back to the passport photo: the LSF of 3 means every length is tripled — but the ratio of any two lengths WITHIN the photo is unchanged. |
| **Lesson 7**<br>**Area Scale Factor — How Does Enlargement Change the Area of the Passport Photo?** | Students drew a 1 × 1 square (object) and a 3 × 3 square (image, LSF = 3) on squared paper, then counted unit squares to find each area. They recorded: Object area = 1 $u^2$, Image area = 9 $u^2$, and computed the ratio of areas. Teacher should then ask students to repeat with a 2 × 4 rectangle (object) and its image under LSF = 3, again counting squares to find the area ratio. Ask: 'What do you predict the area ratio will be for any LSF = 3 enlargement?' Allow 2 minutes wait time before taking responses. | Students derived that the ratio of the image area to the object area equals the SQUARE of the Linear Scale Factor: ASF = $LSF^2$ = $3^2$ = 9. This means the enlarged passport photo requires nine times as much paper. Teacher note: The leap from LSF to ASF is conceptually significant — students must understand WHY it is squared (area is two-dimensional; both dimensions are multiplied by k, so area is multiplied by k × k). Use the grid diagram as the primary explanation, not just the formula. | The teacher generalised: for any two similar figures with linear scale factor k, ASF = $k^2$. Teacher should derive this algebraically for a rectangle (Area = $l \times w \rightarrow kl \times kw = k^2lw$) and then confirm with the square and rectangle examples from the activity. Connect to the passport photo: the photographer printing a 3× enlarged photo uses 9× as much photographic paper — a practical implication for cost and material planning. Ask: 'What would ASF be if the photo were reduced to half size?' (LSF = ½, ASF = ¼). |
| **Lesson 8**<br>**Area Scale Factor: Why Does the Larger Passport Photo Use Nine Times as Much Paper?** | Students were given the areas of the original and enlarged passport photos and several similar polygon pairs (triangles, trapezoids representing farm fields in the Rift Valley). They calculated ASF directly from areas and also from $LSF^2$, then verified both methods gave the same result. Teacher should require students to show BOTH calculation paths side by side in their exercise books. Cold-call four students to explain their reasoning, not just their answers — specifically ask: 'Why is the area ratio not equal to the length ratio?' | Students consolidated ASF = $LSF^2$ = $k^2$ through multiple problem types: (a) given LSF, find ASF; (b) given ASF, find LSF (requiring a square root); (c) given one area and LSF, find the other area. Teacher note: Type (b) is a new skill — finding LSF from ASF requires $\sqrt{(ASF)}$. Prepare students to recognise when to take a square root versus when to square. | The teacher used a visual proof (decomposing the 3× enlarged square into 9 unit squares) alongside the algebraic proof to ensure dual representation. Emphasise the real-world implication: a photographer, fabric printer, or mapmaker working with similar figures must use the ASF — not the LSF — to calculate material quantities and costs. Connect to a Kenyan context: a tailor in Gikomba market scaling up a dress pattern by LSF = 2 will need 4× as much fabric, not 2×. Update DQB: 'We can now explain why enlargement affects area and length differently.' |
| **Lesson 9**<br>**Volume Scale Factor — How Does Enlargement Change the Volume of a 3D Object?** | Students built 1 × 1 × 1 and 3 × 3 × 3 cubes from unit cubes (or sketched them on isometric paper) and counted unit cubes to find each volume. They recorded: Object volume = 1 $u^3$, Image volume = 27 $u^3$, and | Students derived that the Volume Scale Factor (VSF) equals the CUBE of the Linear Scale Factor: VSF = $LSF^3$ = $k^3$. For LSF = 3, VSF = 27 — the enlarged 3D object has 27 times the volume. Teacher note: Connect to | The teacher generalised: for any two similar 3D figures with linear scale factor k, VSF = $k^3$. Derive algebraically for a cuboid ($V = l \times w \times h \rightarrow kl \times kw \times kh = k^3lwh$). Connect to Kenyan contexts: a jua kali artisan in |

| | | | |
|---|---|---|---|
| | computed the ratio of volumes. Teacher should then ask: 'Predict the volume ratio for LSF = 2 and LSF = 4 before calculating.' Allow 3 minutes for group prediction and discussion. Cold-call at least three groups to share predictions and reasoning before confirming with calculation. | ASF: emphasise the pattern $LSF^1$ (length), $LSF^2$ (area), $LSF^3$ (volume) — each dimension adds one power of k. This pattern should be written prominently on the DQB for reference in subsequent lessons. | Kamukunji who scales up a cylindrical water tank by LSF = 2 produces a tank holding 8× as much water — a crucial calculation for rural water storage in arid areas like Turkana. Ask: 'If a model of the JKIA terminal has LSF = 1/100 compared to the real building, what is the VSF?' (VSF = 1/1,000,000). |
| **Lesson 10**<br>**Calculating and Applying the Volume Scale Factor (VSF) to Solve Problems** | Students completed a structured problem set involving 3D similar figures in Kenyan contexts: scaling a cylindrical storage silo near Eldoret, comparing two similar conical piles of maize grain at a Nakuru grain store, and finding unknown volumes of similar spherical water tanks used in Kitui. Teacher should use the 'think-pair-share' protocol: students solve individually for 5 minutes, compare with a partner for 2 minutes, then the teacher cold-calls three pairs to present. Require students to state which scale factor (LSF, ASF, or VSF) they chose and WHY. | Students practised all three VSF calculation types: (a) given LSF, find VSF; (b) given VSF, find LSF (requiring a cube root); (c) given one volume and VSF or LSF, find the other volume. Teacher note: Cube roots are a common stumbling block — ensure students can identify perfect cubes (1, 8, 27, 64, 125) and know how to use a calculator for non-perfect cubes. Also address the diagnostic skill: students must determine from context whether a problem provides a LENGTH ratio, AREA ratio, or VOLUME ratio before selecting the appropriate scale factor formula. | The teacher reinforced the three-tier system: LSF = k, ASF = $k^2$, VSF = $k^3$. Demonstrate the 'working backwards' skill: if VSF = 64, then $k^3 = 64$, so $k = \sqrt[3]{64} = 4$, and ASF = $k^2$ = 16. Work through at least two full examples modelling this chain of reasoning explicitly. Connect to the passport photo: 'The passport photo is 2D, so we only needed LSF and ASF — but a 3D version of this problem (e.g., a sculptor scaling a bust) would also require VSF.' Update DQB. |
| **Lesson 11**<br>**EXPLAIN & APPLY: Mixed Problems — LSF, ASF, and VSF** | Students worked on a mixed problem set containing 8 problems requiring them to first diagnose which scale factor is relevant, then execute the correct calculation. Problems were drawn from diverse Kenyan contexts: map scales for Nairobi County, fabric areas for a Maasai shuka pattern, volumes of similar clay pots from a Kisii soapstone carver, and similar triangles on a surveyor's plan for a Mombasa coastal plot. Teacher should use a 'gallery walk' structure: groups post their solutions on the wall; other groups use sticky notes to flag errors or offer alternative approaches. Allow 15 minutes for the gallery walk. | Students consolidated that LSF, ASF (= $LSF^2$), and VSF (= $LSF^3$) form a single unified system. The critical meta-skill is diagnosis: identifying whether a problem provides a length ratio, area ratio, or volume ratio before selecting the correct formula. Students also learned to chain calculations (e.g., find LSF from an area ratio, then use LSF to find a missing length). Teacher note: The gallery walk typically reveals 2–3 recurring error types; address these whole-class before the lesson close. | The teacher conducted a whole-class debrief of the gallery walk, targeting the most common errors identified on sticky notes. Formalise the diagnostic decision tree on the board: 'Given a LENGTH ratio → that IS the LSF. Given an AREA ratio → $\sqrt{\text{area ratio}}$ = LSF. Given a VOLUME ratio → $\sqrt[3]{\text{volume ratio}}$ = LSF. Then apply whichever formula the problem requires.' Connect to the driving question: 'The passport photo problem is a length and area problem — we now have all the tools to answer it completely.' Update DQB with remaining questions. |
| **Lesson 12**<br>**Final Explanation — Weaving the Three Relationships: LSF, ASF, and VSF** | Students returned to the original passport photo phenomenon from Lesson 1 and answered a structured set of final questions: (1) What is the LSF between the two passport photos? (2) How much more paper does the large photo require? (3) If the photos were 3D objects, what would the volume ratio be? (4) Why is the shape of | Students consolidated the complete conceptual architecture of the sub-strand: ASF = $LSF^2$ and VSF = $LSF^3$, meaning a single Linear Scale Factor simultaneously determines how lengths, areas, and volumes change between similar figures. All three scale factors are unified by the dimension of the measurement (1D → k, 2D | The teacher delivered the final synthesis: the mathematics that makes resizing a passport photo possible is the Linear Scale Factor — a single multiplier applied uniformly to every length. Because every length is scaled by the same factor k, all angles remain unchanged (shape is preserved), all areas scale by $k^2$ (hence the |

| | | | |
|---|---|---|---|
| | the face preserved even though the size changes? Students compared their Lesson 12 answers with their Lesson 1 predictions on the DQB. Teacher should facilitate a DQB close-out: for each original student question, ask 'Can we answer this now? How?' Celebrate the growth in understanding explicitly. | $\rightarrow k^2$, 3D $\rightarrow k^3$). Shape is preserved because ALL lengths are multiplied by the SAME factor, keeping all angle ratios and side ratios constant. Teacher note: This is the lesson to address any residual misconceptions from the DQB. Ensure the driving question is answered explicitly and in full sentences, not just with formulas. | photographer uses more paper in a predictable ratio), and all volumes would scale by $k^3$. The teacher should close by asking students to write a one-paragraph answer to the driving question in their own words — this serves as both a formative assessment and a celebration of the full learning journey. Connect to career relevance: architects, mapmakers, photographers, engineers, and tailors across Kenya use exactly this mathematics every day. |